

CAMERON CRADDOCK

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<https://www.linkedin.com/in/cameron-craddock/> | <https://github.com/ccraddock>

Multi-disciplined engineer with expertise in productizing AI, medical physics, computational radiology, and managing large-scale imaging and behavioral data collection studies

TECHNICAL SKILLS

Pytorch, Python, C/C++, Matlab, R, BASH, Pandas, machine learning (supervised/unsupervised), data science, multivariate statistics, Monte Carlo methods, image/signal processing, embedded systems, HPC, cloud computing, mobile applications, containers, CI/CD, version control, real-time OS, Linux, networking, biomedical research, electromyogram (EMG), medical imaging (MRI, fMRI, PET, DICOM)

PROFESSIONAL EXPERIENCE

Staff Machine Learning Engineer (IC6), Meta Platforms Inc

Remote | 08/2022 - Present

Tech Lead on a team productizing AI algorithms for a low-power, EMG-based Brain-Computer Interface (Meta Neural Band). Implemented on-device inference on 16-bit NPUs with performance parity to floating-point models.

- Designed, evaluated and implemented new features in model architecture and training to optimize accuracy and system performance for three EMG time-course gesture decoder models (CNN, LSTM).
- Resolved model NPU incompatibilities by replacing unsupported components with functional equivalents.
- Implemented pytorch training and evaluation pipelines running on multi-gpu cloud infrastructure.
- Executed biweekly model release process from branch cut to firmware packaging and helped implement a push on green pipeline to fully automate the process.
- Designed and executed A/B user studies to measure model performance on-device leveraging power analyses to maximize study efficiency.
- Implemented data quality assessment, control, and labeling pipelines and data monitoring dashboards.
- Refactored multi-sensor fusion feature pipeline to achieve a 10x improvement in temporal alignment.
- Implemented numerically accurate and power optimized preprocessing using various DSP techniques.
- Lead a cross-organizational team to build an automated high-resolution power measurement system with dashboards for monitoring inference pipeline performance and quickly mitigating regressions.
- Lead the development of tools to estimate PyTorch model performance (power/latency) on various NPU configurations, enabling power and latency -informed Neural Architecture Search (NAS).

Research Engineering Manager, Indeed

Austin, Texas | 8/2021 - 8/2022

- Tech Lead/Manager position supporting six engineers to optimize job posting placement in [Career Guide](#).
- Designed a bag-of-words based clustering system for organizing articles into subgroups for pooling statistics to interpolate reader profiles for new articles.

Associate Professor of Diagnostic Medicine, The University of Texas at Austin Dell Medical School

Austin, Texas | 9/2017 - 07/2021

Specialized in Radiology within Diagnostic Medicine, which encompasses both Radiology and Pathology.

- Managed a federal grant-supported research agenda developing AI algorithms to predict diagnosis, prognosis, and treatment outcomes from neuroimaging data for a variety of neuropsychiatric conditions.
- Led the development of a mobile phone app for collecting behavioral and questionnaire data for measuring the impact of environmental exposures on mental health.

- Developed fMRI-based brain computer interfaces for exploring network modulation and neurofeedback.
- Developed [C-PAC](#), an open source neuroimaging software for performing high-throughput image processing using cloud and HPC systems.
- Collaborated across the university to optimize MR imaging protocols for multi-modal brain characterization, pancreas quantification and whole body fat quantization experiments.
- Developed cloud based neuroimaging AI pipelines to support fast turn-around clinical assessments.
- Designed and executed large scale longitudinal data collection efforts with whole body imaging with neuropsychiatric assessment.

Director of the Computational Neuroimaging Lab, Nathan S. Kline Institute for Psychiatric Research & Director of Imaging, Child Mind Institute

Orangeburg, NY | 6/2012 - 8/2017

- Managed a software engineering and data science team to develop novel MRI data acquisition and analysis tools for elucidating mechanisms underlying neuropsychiatric disorders.
- Managed a neuroimaging data collection infrastructure that collected and openly shared hundreds of MRI scans a month as a part of the [Healthy Brain Network](#):
 - Optimized MRI sequences for functional, structural and molecular neuroimaging, and abdominal imaging in collaboration with Siemens research scientists.
 - Monitored image quality and worked with service engineers to resolve issues as they arose.
 - Built PACS for image collection and transfer to long-term storage and for radiological reads.
 - Directly oversaw operations of a Siemens 1.5T Avanto mobile MRI, including training operators, defining study procedures, and overseeing safety protocols.
 - Established and managed image acquisition on Siemens 3T Tim Trio systems at partner sites.
 - Coordinated radiological reads.

OPEN SCIENCE

- Co-founder of [Brainhack](#), an open science organization that organizes hackathons to facilitate interdisciplinary collaborative approaches to neuroscience; over 150 events have been held since 2012.
- Founded [Preprocessed Connetomes Project](#) for sharing preprocessed neuroimaging data.
- Founding Chair of the Organization for Human Brain Mapping Open Science SIG (2016).
- Served as Education Chair for the Organization for Human Brain Mapping (2017 - 2019).

GRANTS, AWARDS AND PUBLICATIONS

- Author on over 98 peer reviewed scientific publications and a patent ([h-index: 58, i-10 index: 98](#)).
- Invited speaker at over 35 conferences, workshops, and symposia including SXSW (2016 & 2017).
- 2021 Organization for Human Brain Mapping Open Science Award (\$2.5K cash prize).
- 2014 NIMH Biobehavioral Research Award for Innovative New Researchers (\$2.6MM grant).
- 2010 NARSAD Young Investigator Grant (\$75K grant).

EDUCATION

PhD in Electrical and Computer Engineering (Bioengineering and AI)

Dissertation Title: Disease state prediction from resting state functional connectivity

Georgia Institute of Technology, Atlanta, GA | 2009

Bachelor of Computer Engineering

Georgia Institute of Technology, Atlanta, GA | 1999